

Solutions – Quiz 1

ECON 300: Labour Economics

Instructor: Muhebullah Karimzada

January 27, 2026

Multiple Choice (5 points)

1. **Answer: (c).** A discouraged worker who stopped searching is counted as **not in the labour force** because they are not actively searching for work. In standard labour force concepts, active search is required to be classified as unemployed.
2. **Answer: (b).** The employment-to-population ratio is:

$$\frac{E}{\text{Working-age population}}.$$

It measures the share of the working-age population that is employed, regardless of whether others are unemployed or out of the labour force.

3. **Answer: (c).** An increase in non-labour income raises consumption at every leisure choice without changing the wage. Graphically, it shifts the budget line upward in a parallel way in the labour-leisure diagram.
4. **Answer: (c).** At sufficiently high wages, the income effect can dominate the substitution effect, leading people to choose more leisure and fewer work hours. That generates a backward-bending labour supply curve over that range.
5. **Answer: (b).** A more generous unemployment insurance system can lengthen search duration or reduce the urgency of accepting a job, and can also affect measured participation through search behaviour, potentially lowering measured participation if some people reduce active search or delay re-entry. (The direction can depend on program design, but among the options, this is the best fit.)

Problem 1: Labour Force Measurement and Discouragement (7.5 points)

Given initial conditions:

$$\text{Working-age population} = 900,000, \quad E = 585,000, \quad U = 65,000, \quad N = 250,000.$$

(a) Labour force, LFPR, unemployment rate (2.5 points)

Step 1: Labour force.

$$LF = E + U = 585,000 + 65,000 = 650,000.$$

Step 2: LFPR.

$$LFPR = \frac{LF}{\text{Working-age population}} = \frac{650,000}{900,000} = 0.7222 \dots \approx 72.22\%.$$

Step 3: Unemployment rate.

$$u = \frac{U}{LF} = \frac{65,000}{650,000} = 0.10 = 10\%.$$

(b) Employment-to-population ratio and interpretation (2.5 points)

Step 1: Employment-to-population ratio.

$$\frac{E}{\text{Working-age population}} = \frac{585,000}{900,000} = 0.65 = 65\%.$$

Step 2: Conceptual difference from LFPR. The employment-to-population ratio measures the share of the working-age population that is actually employed. The LFPR measures the share of the working-age population that is *engaged with the labour market*, meaning employed or actively searching (unemployed). Therefore, LFPR can be high even if employment is not high, as long as many people are searching (unemployed).

(c) Slowdown: job losses and discouragement (2.5 points)

Event 1: 15,000 employed lose jobs. Assume they initially move to unemployment (they want work).

$$E' = 585,000 - 15,000 = 570,000, \quad U' = 65,000 + 15,000 = 80,000.$$

Event 2: 10,000 unemployed stop searching. They move from unemployed to not in the labour force.

$$U'' = 80,000 - 10,000 = 70,000.$$

Now the labour force is:

$$LF'' = E' + U'' = 570,000 + 70,000 = 640,000.$$

New unemployment rate:

$$u'' = \frac{U''}{LF''} = \frac{70,000}{640,000} = 0.109375 \approx 10.94\%.$$

Explanation. Discouragement reduces measured unemployment because individuals who stop searching are no longer counted as unemployed; they exit the labour force. This changes both:

- the numerator U (falls when people stop searching), and
- the denominator LF (also falls when people stop searching).

As a result, the unemployment rate may not rise as much as the true deterioration in job opportunities would suggest. This is why economists also examine participation rates and employment-to-population ratios when evaluating labour market conditions.

Problem 2: Labour Supply and Minimum Wage (7.5 points)

Given individual decision:

$$C = 500 + 30h, \quad U = C - h^2.$$

Given market:

$$L^D = 150 - 3w, \quad L^S = 30 + 2w.$$

(a) Optimal hours of work (2.5 points)

Step 1: Substitute the budget constraint into utility.

$$U(h) = (500 + 30h) - h^2 = 500 + 30h - h^2.$$

Step 2: Take the derivative and set to zero (FOC).

$$\frac{dU}{dh} = 30 - 2h.$$

FOC:

$$30 - 2h = 0 \quad \Rightarrow \quad 2h = 30 \quad \Rightarrow \quad h^* = 15.$$

Step 3: Second-order condition.

$$\frac{d^2U}{dh^2} = -2 < 0,$$

so the solution is a maximum.

Answer: The worker supplies **15 hours per week**.

(b) Competitive equilibrium wage and employment (2.5 points)

Step 1: Set $L^D = L^S$.

$$150 - 3w = 30 + 2w.$$

Step 2: Solve for w .

$$150 - 30 = 2w + 3w \quad \Rightarrow \quad 120 = 5w \quad \Rightarrow \quad w^* = 24.$$

Step 3: Compute equilibrium employment.

$$L^* = 150 - 3(24) = 150 - 72 = 78.$$

(Check: $30 + 2(24) = 30 + 48 = 78$.)

Answer: Competitive equilibrium is $w^* = 24$ and $L^* = 78$.

(c) Minimum wage $w = 25$ (2.5 points)

Step 1: Is it binding? Compare to equilibrium:

$$w_{\min} = 25 > w^* = 24,$$

so the minimum wage is **binding**.

Step 2: Labour demanded at $w = 25$.

$$L^D(25) = 150 - 3(25) = 150 - 75 = 75.$$

Step 3: Labour supplied at $w = 25$.

$$L^S(25) = 30 + 2(25) = 30 + 50 = 80.$$

Step 4: Unemployment (excess supply).

$$U = L^S - L^D = 80 - 75 = 5.$$

Answer: The minimum wage is **binding** and generates **5** units of unemployment (in the same units as L , e.g., thousands if L is in thousands).